

# Daniil Dmitriev

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| ACADEMIC POSITIONS | <b>University of Pennsylvania</b><br>Postdoctoral Researcher<br>Department of Statistics and Data Science, The Wharton School<br>Hosted by Prof. Yuting Wei.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Sep. 2025–Present                    |
| EDUCATION          | <b>ETH Zurich</b><br>Ph.D., Department of Mathematics<br>Advisors: Prof. Afonso S. Bandeira and Prof. Fanny Yang.<br>Thesis: <i>High-Dimensional Optimization Problems in Average-Case and Worst-Case Settings</i> .<br><b>École Polytechnique Fédérale de Lausanne</b><br>M.Sc. in Data Science<br><b>Moscow Institute of Physics and Technology</b><br>B.Sc. in Applied Mathematics and Computer Science                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2025<br><br><br><br>2021<br><br>2018 |
| RESEARCH INTERESTS | High-dimensional statistics, statistical learning theory, and mathematical foundations of generative models. I work on understanding how the intrinsic structure affects learning and sampling in high dimensions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                      |
| PREPRINTS          | [1] <b>Daniil Dmitriev</b> , Z. Huang, and Y. Wei. “Provably adaptive sampling with uniform and remasking discrete diffusion models.” Under review at <i>Annals of Applied Probability</i> , 2026.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                      |
| PUBLICATIONS       | [1] <b>Daniil Dmitriev</b> <sup>*</sup> , Z. Huang <sup>*</sup> , and Y. Wei. “Efficient sampling with discrete diffusion models: sharp and adaptive guarantees.” <i>Conference on Learning Theory (COLT)</i> , 2026.<br>[2] <b>Daniil Dmitriev</b> , H. Franck, C. Heinzler, and A. Sanyal. “Learning in an echo chamber: online learning with replay adversary.” <i>ACM–SIAM Symposium on Discrete Algorithms (SODA)</i> , 2026. Selected for FORC Highlights, 2026.( $\alpha\beta$ )<br>[3] A. Bandeira, J. Błasiok, <b>Daniil Dmitriev</b> , U. Faure, A. Kireeva, and D. Kunisky. “The Lovász number of random circulant graphs.” <i>Sampling Theory and Applications (SampTA)</i> , 2025.( $\alpha\beta$ )<br>[4] <b>Daniil Dmitriev</b> <sup>*</sup> , R.-D. Buhai <sup>*</sup> , S. Tiegel, A. Wolters, G. Novikov, A. Sanyal, D. Steurer, and F. Yang. “Robust mixture learning when outliers overwhelm small groups.” <i>Advances in Neural Information Processing Systems (NeurIPS)</i> , 2024.<br>[5] <b>Daniil Dmitriev</b> , K. Szabó, and A. Sanyal. “On the growth of mistakes in differentially private online learning: a lower bound perspective.” <i>Conference on Learning Theory (COLT)</i> , 2024.<br>[6] D. Schröder <sup>*</sup> , <b>Daniil Dmitriev</b> <sup>*</sup> , H. Cui <sup>*</sup> , and B. Loureiro. “Asymptotics of learning with deep structured (random) features.” <i>International Conference on Machine Learning (ICML)</i> , 2024.<br>[7] G. Arpino, <b>Daniil Dmitriev</b> , and N. Grometto. “Greedy heuristics and linear relaxations for the random hitting set problem.” <i>International Conference on Approximation Algorithms for Combinatorial Optimization Problems (APPROX)</i> , 2024.( $\alpha\beta$ )<br>[8] D. Schröder, H. Cui, <b>Daniil Dmitriev</b> , and B. Loureiro. “Deterministic equivalent and error universality of deep random features learning.” <i>International Conference on Machine Learning (ICML)</i> , 2023.<br>[9] <b>Daniil Dmitriev</b> and M. Zhukovskii. “On monotonicity of Ramanujan function for binomial random variables.” <i>Statistics &amp; Probability Letters</i> , 2021.( $\alpha\beta$ )<br>[10] T. Lin, S. U. Stich, L. Barba, <b>Daniil Dmitriev</b> , and M. Jaggi. “Dynamic model pruning with feedback.” <i>International Conference on Learning Representations (ICLR)</i> , 2020. |                                      |

- [11] **Daniil Dmitriev** and M. Zhukovskii. “On a connection of two theoretical graph problems with conjectures of Ramanujan and Samuels.” *Russian Mathematical Surveys*, 2018. ( $\alpha\beta$ )  
 \*Equal contribution. ( $\alpha\beta$ ): Alphabetical order

**AWARDS** **ETH AI Center Fellowship** (top 2% of applicants) 2021–2025  
**COLT Student Travel Award** 2024  
**NeurIPS 2024 Top Reviewer** 2024

**SELECTED TALKS** IMS Annual Meeting, Invited talk Salzburg, 2026  
*Efficient sampling with discrete diffusion models: sharp and adaptive guarantees*  
 Learning-driven Algorithms and Machine-aided Proofs (LAMP), TTIC, 2026  
 Spotlight talk  
*Learning in an echo chamber: online learning with replay adversary*  
 Conference on Learning Theory San Diego, 2026  
*Efficient sampling with discrete diffusion models: sharp and adaptive guarantees*  
 Foundations of Responsible Computing (FORC), Contributed talk Harvard, 2026  
*Learning in an echo chamber: online learning with replay adversary*  
 Theory Seminar University of Pennsylvania, 2026  
*Two problems in online learning*  
 Youth in High Dimensions, Invited talk ICTP, 2025  
*The Lovász number of random circulant graphs*  
 ETH AI Center Seminar ETH Zurich, 2025  
*Robust Mixture Learning when Outliers Overwhelm Small Groups*  
 Jason Altschuler’s research group University of Pennsylvania, 2024  
*Robust Mixture Learning when Outliers Overwhelm Small Groups*  
 DACO Seminar ETH Zurich, 2024  
*Robust Mixture Learning when Outliers Overwhelm Small Groups*  
 Conference on Learning Theory Edmonton, 2024  
*On the growth of mistakes in differentially private online learning: a lower bound perspective*  
 Youth in High Dimensions, Contributed talk ICTP, 2024  
*Robust Mixture Learning when Outliers Overwhelm Small Groups*  
 Graduate seminar in probability ETH Zurich, 2024  
*Greedy heuristics and linear relaxations for the random hitting set problem*  
 DeLTA Seminar University of Copenhagen, 2024  
*Lower bounds for private online learning*  
 Workshop on Spin Glasses, Invited talk Les Diablerets, 2022  
*Integrality gaps for vertex covers on sparse Bernoulli hypergraphs*  
 ETH AI Center Doctoral Seminar ETH Zurich, 2022  
*Random matrices for deep learning theory*

**RESEARCH INTERNSHIPS AND VISITS** **Visiting Graduate Student**, Simons Institute for the Theory of Computing UC Berkeley, 11/2024-12/2024  
*Program: Modern Paradigms in Generalization.*  
**Research Scholar Student**, MLO Lab EPFL, 2018-2020  
*Working on neural network pruning. Host: Prof. Martin Jaggi*  
**Research AI Intern**, Bloomberg London, 09/2020-01/2021  
*Worked on diversification techniques for recommendation system.*

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|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| TEACHING         | <b>ETH Zurich:</b> Introduction to Machine Learning                                                                                                                                                                                                                                  | Spring 2025       |
|                  | <i>Teaching assistant: homework and exercise session preparation</i>                                                                                                                                                                                                                 |                   |
|                  | Mathematics of Machine Learning                                                                                                                                                                                                                                                      | Spring 2022       |
|                  | <i>Teaching assistant: leading exercise sessions and contributing to exam questions</i>                                                                                                                                                                                              |                   |
|                  | Mathematics of Data Science                                                                                                                                                                                                                                                          | Fall 2021         |
|                  | <i>Teaching assistant: leading exercise sessions and contributing to exam questions</i>                                                                                                                                                                                              |                   |
| MENTORING        | <b>EPFL:</b> Artificial Neural Networks                                                                                                                                                                                                                                              | Spring 2021       |
|                  | <i>Teaching assistant: leading exercise sessions and homework preparation</i>                                                                                                                                                                                                        |                   |
|                  | Artificial Neural Networks                                                                                                                                                                                                                                                           | Spring 2020       |
|                  | <i>Teaching assistant: leading exercise sessions and homework preparation</i>                                                                                                                                                                                                        |                   |
|                  | Since 2022, I have mentored seven Master's, Bachelor's, and semester project students as a primary supervisor. These collaborations have led to publications in prestigious venues (SODA, COLT, SampTA), and mentees now pursue Ph.D. programs at University of Copenhagen and ISTA. |                   |
|                  | <b>M.Sc. Theses, ETH Zurich:</b>                                                                                                                                                                                                                                                     |                   |
| ACADEMIC SERVICE | Ulysse Faure                                                                                                                                                                                                                                                                         | Spring 2024       |
|                  | Krish Agrawal                                                                                                                                                                                                                                                                        | Spring 2024       |
|                  | Carolin Heinzler → Ph.D. student at University of Copenhagen                                                                                                                                                                                                                         | Fall 2023         |
|                  | <b>B.Sc. Theses, ETH Zurich:</b>                                                                                                                                                                                                                                                     |                   |
|                  | Emanuele Barbieri → M.Sc. at ETH Zurich                                                                                                                                                                                                                                              | Spring 2023       |
|                  | Minerva Snellman → M.Sc. at ETH Zurich                                                                                                                                                                                                                                               | Spring 2022       |
|                  | <b>Semester Projects, ETH Zurich:</b>                                                                                                                                                                                                                                                |                   |
|                  | Filip Kovačević → Ph.D. student at ISTA                                                                                                                                                                                                                                              | Spring 2023       |
|                  | Kristóf Szabó                                                                                                                                                                                                                                                                        | Spring 2023       |
|                  | <b>Conference reviewing:</b> NeurIPS 2024 (Top Reviewer); ICLR 2025; ICML 2025; COLT 2025; ICLR 2026; AISTATS 2026; COLT 2026.                                                                                                                                                       |                   |
|                  | <b>Journal reviewing:</b> TMLR; <i>SIAM Journal on Mathematics of Data Science</i> ; JMLR.                                                                                                                                                                                           |                   |
|                  | <b>Organizer:</b> Theory Breakfast, ETH Zurich                                                                                                                                                                                                                                       | Spring 2024, 2025 |
|                  | <i>Weekly event for PhD students and postdocs across mathematics, optimization, and theoretical computer science groups.</i>                                                                                                                                                         |                   |